

MUCKLE HOLM, United Kingdom

WMO: 030070

Lat: 60.581N

Lon: 1.265W

Elev: 20

StdP: 101.08

Time Zone: 0.00 (EUW)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.7	0.2	-6.4	2.2	1.5	-4.9	2.5	1.8	24.7	5.9	22.8	4.7	9.0	320	0.868

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.5	15.6	14.0	14.7	13.3	14.1	12.8	14.4	15.2	13.7	14.5	13.1	13.9	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	14.9	13.3	9.5	14.3	12.7	9.1	13.7	40.4	15.4	38.6	14.5	37.1	14.0	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.3	15.7	DB	-2.7	17.8	2.8	1.2	-4.7	18.7	-6.3	19.3	-7.9	20.0	-10.0	20.8	(4)
(5)			WB	-4.0	15.8	2.3	1.5	-5.6	16.8	-7.0	17.7	-8.3	18.5	-10.0	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	5.6	4.9	5.2	6.0	7.2	9.4	11.4	12.6	11.2	9.0	7.2	5.6	(6)	
	DBStd	3.24	1.81	2.15	1.98	1.83	2.76	1.74	1.77	1.27	1.56	1.90	1.89	2.23	(7)	
	HDD10.0	944	137	144	148	120	88	30	7	0	7	40	86	137	(8)	
	HDD18.3	3786	396	377	407	370	344	268	214	177	215	288	335	396	(9)	
	CDD10.0	201	0	0	0	0	3	12	51	82	42	10	1	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.3	8.8	9.6	8.4	7.1	5.7	6.0	5.2	5.2	6.6	8.3	8.5	8.0	(14)	
(15) Precipitation	PrecAvg	1154	145	113	106	64	59	57	60	82	91	115	130	133	(15)	
	PrecMax	1395	278	174	186	107	123	109	131	160	140	182	195	233	(16)	
	PrecMin	980	46	51	39	22	9	10	22	38	28	47	58	51	(17)	
	PrecStd	117	52	30	28	18	24	24	25	28	28	33	33	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.2	9.9	10.8	14.7	15.3	16.8	17.0	15.6	13.1	11.7	10.9	(19)	
		MCWB	8.4	9.3	8.6	9.0	11.3	12.0	14.9	15.5	14.0	12.2	10.8	9.9	(20)	
	2%	DB	9.2	9.5	9.1	9.7	12.1	13.1	15.2	15.9	14.4	12.3	10.8	10.0	(21)	
		MCWB	8.0	8.5	8.0	8.4	10.2	11.0	13.7	14.5	13.2	11.3	9.9	9.2	(22)	
	5%	DB	8.6	8.6	8.5	9.0	10.8	12.1	14.3	15.0	13.7	11.9	10.3	9.3	(23)	
		MCWB	7.6	7.5	7.4	8.0	9.4	10.6	13.1	13.7	12.6	10.9	9.4	8.4	(24)	
	10%	DB	8.1	7.9	7.9	8.3	9.9	11.5	13.6	14.3	13.2	11.5	9.8	8.7	(25)	
		MCWB	7.1	6.8	6.8	7.2	8.7	10.2	12.5	13.1	12.1	10.5	8.9	7.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.7	9.2	9.7	11.9	12.5	15.2	15.9	14.3	12.6	11.1	10.2	(27)	
		MCDB	9.4	9.9	9.6	10.2	14.1	14.0	16.5	16.7	14.9	12.9	11.5	10.5	(28)	
	2%	WB	8.4	8.7	8.3	8.9	10.5	11.6	14.1	14.7	13.6	11.7	10.3	9.4	(29)	
		MCDB	8.8	9.3	8.7	9.5	11.8	12.7	14.9	15.5	14.2	12.1	10.6	9.9	(30)	
	5%	WB	7.9	7.7	7.7	8.1	9.8	11.1	13.4	14.1	13.0	11.2	9.8	8.8	(31)	
		MCDB	8.4	8.3	8.2	8.6	10.7	11.8	14.1	14.8	13.5	11.7	10.2	9.3	(32)	
	10%	WB	7.3	7.0	7.0	7.5	9.1	10.6	12.7	13.4	12.3	10.7	9.1	7.9	(33)	
		MCDB	8.0	7.8	7.7	8.0	9.7	11.3	13.4	14.2	12.9	11.4	9.7	8.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.5	2.8	2.8	2.7	2.5	2.3	2.4	2.5	2.2	2.2	2.4	2.7	(35)	
		MCDBR	2.9	3.4	3.2	3.4	3.6	3.2	3.5	3.5	2.8	2.3	2.9	2.8	(36)	
	5% WB	MCWBR	3.2	3.5	3.2	2.9	3.1	2.7	2.9	3.0	2.4	2.5	3.4	3.0	(37)	
		MCWBR	3.0	3.5	3.2	3.1	3.4	3.0	3.6	3.4	2.7	2.5	2.9	2.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.355	0.369	0.383	0.373	0.366	0.379	0.366	0.363	0.346	0.328	0.407	(40)	
		taud	2.075	2.194	2.269	2.298	2.394	2.455	2.434	2.471	2.437	2.317	2.130	2.376	(41)	
	Edn at Noon		368	607	744	810	852	865	841	825	755	619	374	241	(42)	
		Edh at Noon	47	73	92	107	105	101	101	90	79	64	43	31	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.22	0.71	1.78	3.34	4.66	4.84	4.36	3.54	2.14	1.07	0.32	0.14	(44)	
	RadStd		0.02	0.08	0.13	0.19	0.30	0.52	0.38	0.34	0.20	0.11	0.03	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)		+0.35	+0.67	+0.70	N/A	N/A	N/A	-102	-134	N/A	N/A

Nomenclature: See separate page